

INTRODUCTION TO AI

Evolution of AI from Simple Concepts

Evolution of AI

- 1950s: Birth of AI
- Machine Learning
- Deep Learning
- Generative AI



Types of AI

NARROW AI (WEAK AI)



Specialized in
one task only

Example: Siri,
Google Translate

GENERAL AI (STRONG AI)



Can think, reason
& learn like humans

Example:
Still under research

SUPER AI



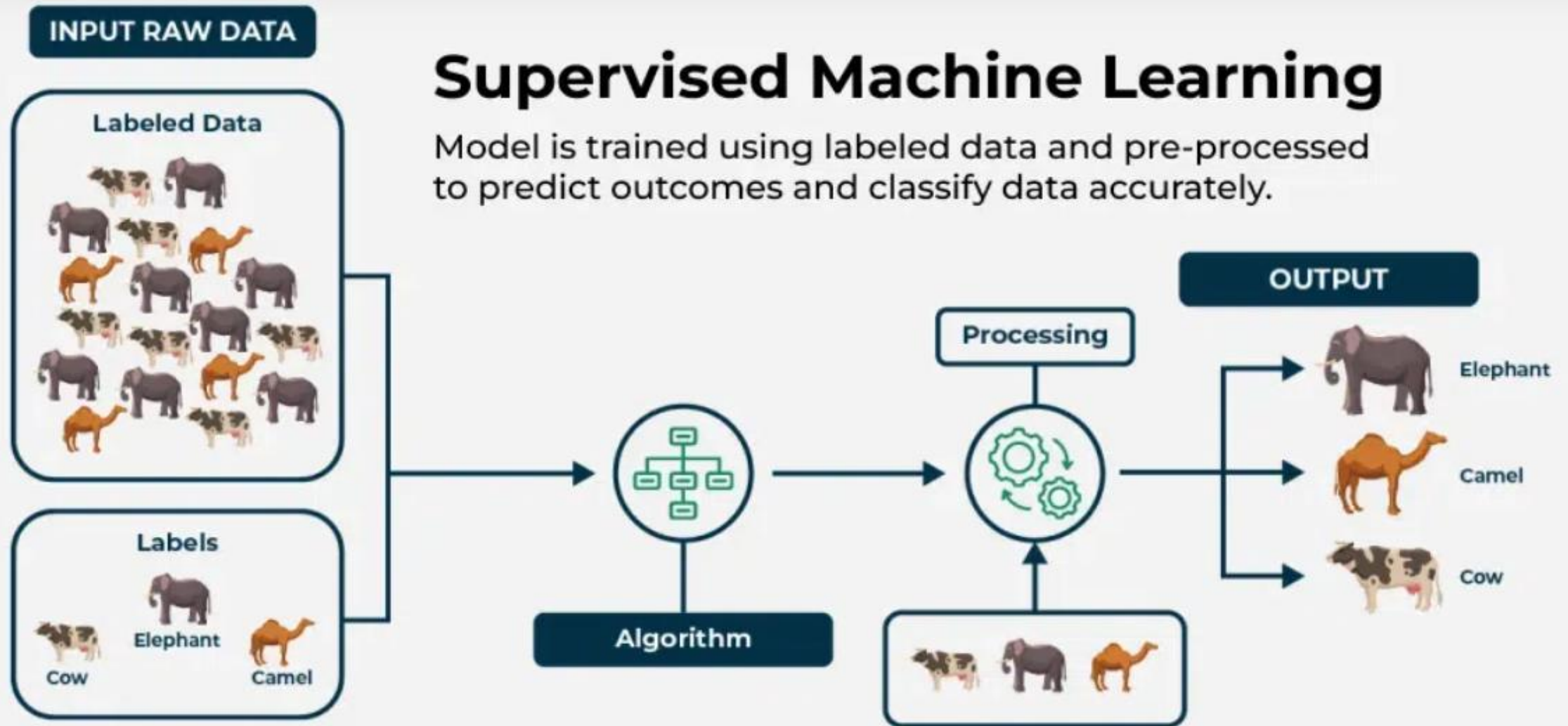
Surpasses human
intelligence

Example:
Theoretical / Future

Supervised Learning

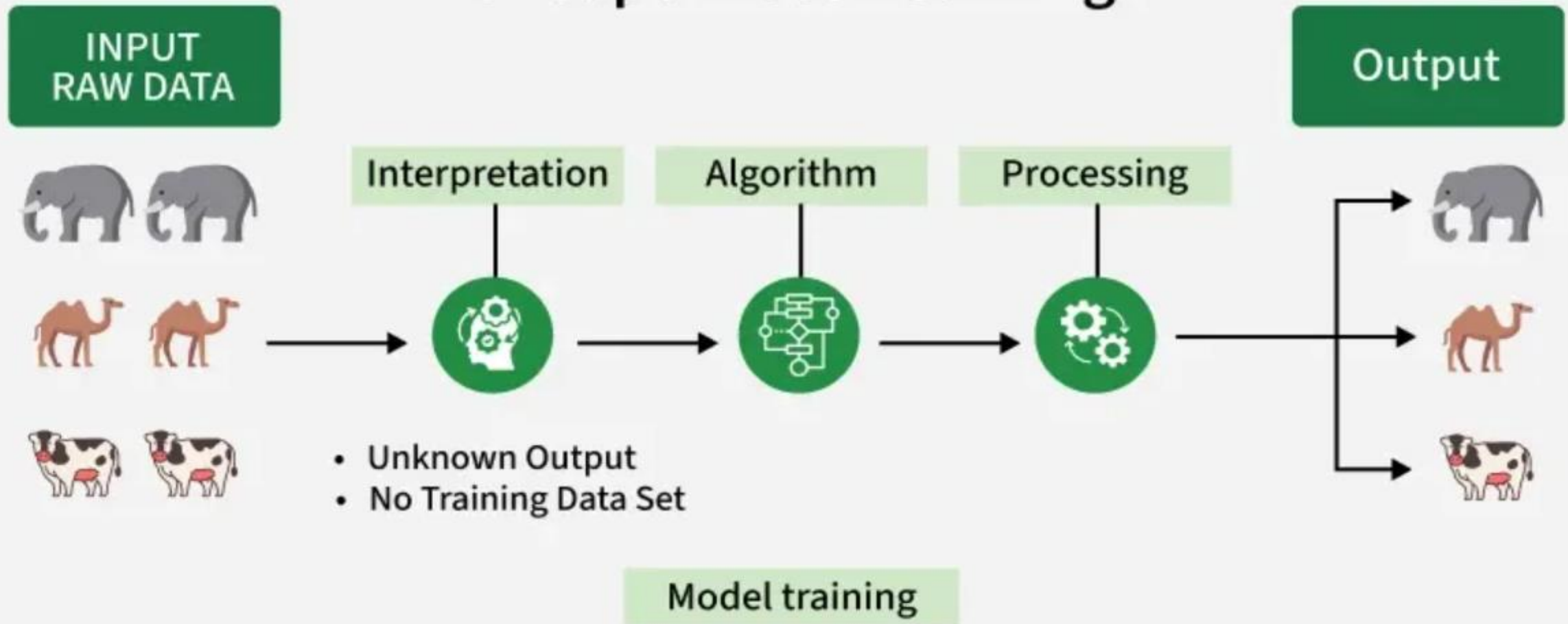
Supervised Machine Learning

Model is trained using labeled data and pre-processed to predict outcomes and classify data accurately.



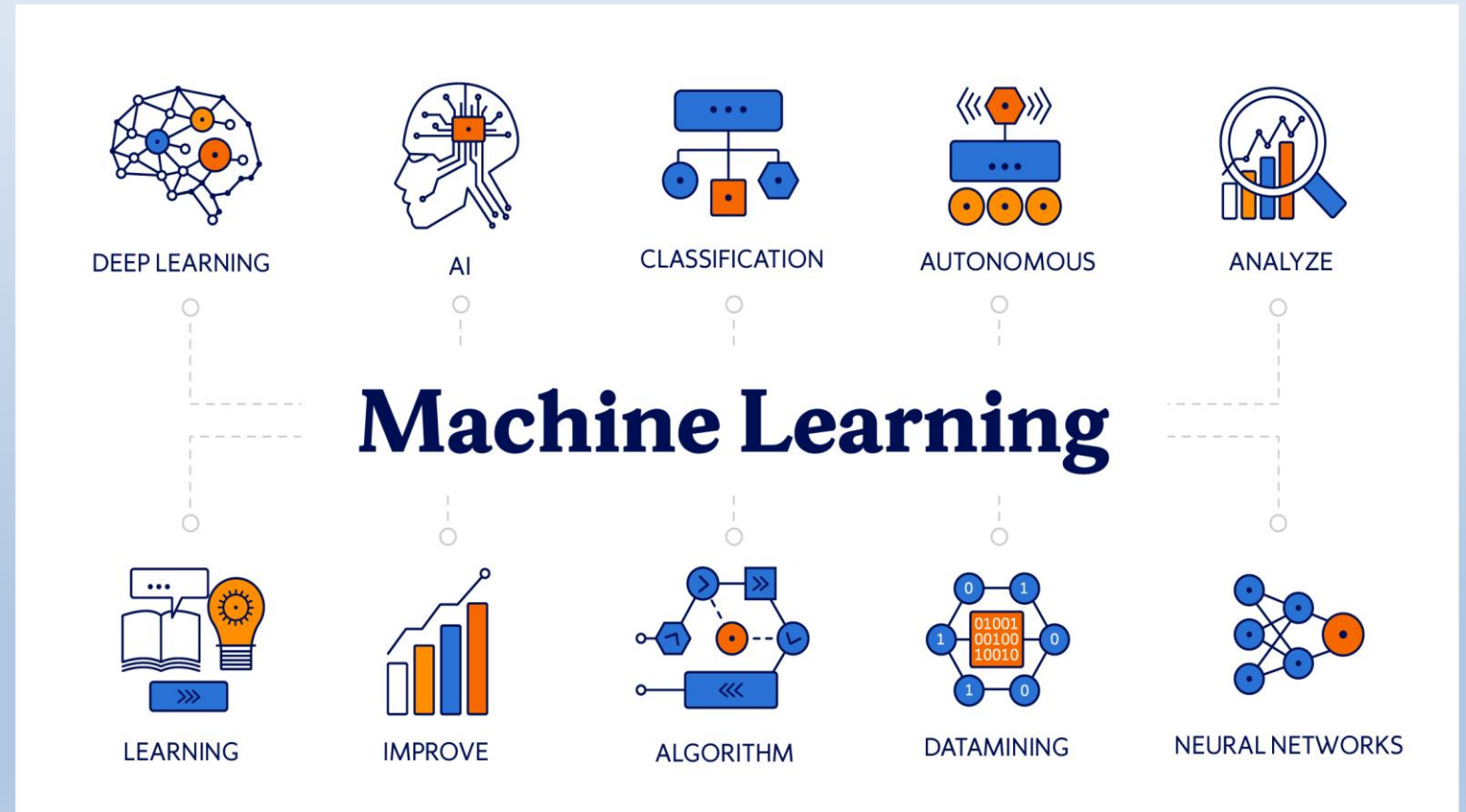
Unsupervised Learning

Unsupervised Learning



What is Machine Learning ?

- Learns from Data
- Finds Patterns
- Makes Predictions
- Improves Accuracy
- Automates Tasks
- Solves Problems



Guess What is the Core of AI?

Artificial Intelligence (AI)

Can AI think by itself?

What is the basic function of AI?



Human Brain ?



Mathematics?

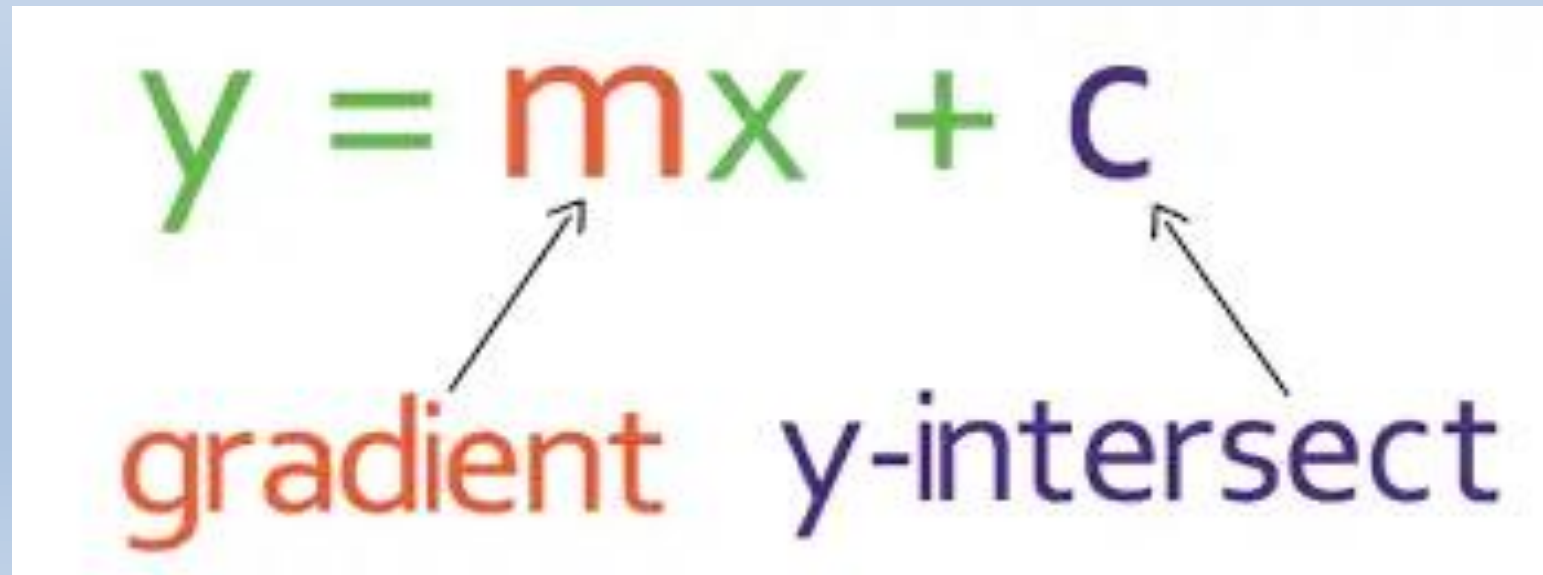


Equation?

Let's Find Out... →

Linear Regression

- Equation: $y = mx + c$
- Predicts continuous values



The diagram shows the linear regression equation $y = mx + c$ in a large font. Below the equation, the word "gradient" is written in orange, and the word "y-intersect" is written in dark blue. An arrow points from "gradient" to the variable m in the equation, and another arrow points from "y-intersect" to the variable c .

$$y = mx + c$$

gradient y-intersect

Linear Regression Example

$$y = mx + c$$

$$X=[1,2,3,4,5] \quad Y=[3,5,7,9,11]$$

$$X_{\text{mean}}=1+2+3+4+5=3, y_{\text{mean}}=3+5+7+9+11=7$$

Numerator:

$$\begin{aligned} \sum(X-X_{\text{mean}}) \cdot (y-y_{\text{mean}}) &= (1-3)(3-7) + (2-3)(5-7) + \\ &\quad (3-3)(7-7) + (4-3)(9-7) + (5-3)(11-7) = 20 \end{aligned}$$

Denominator:

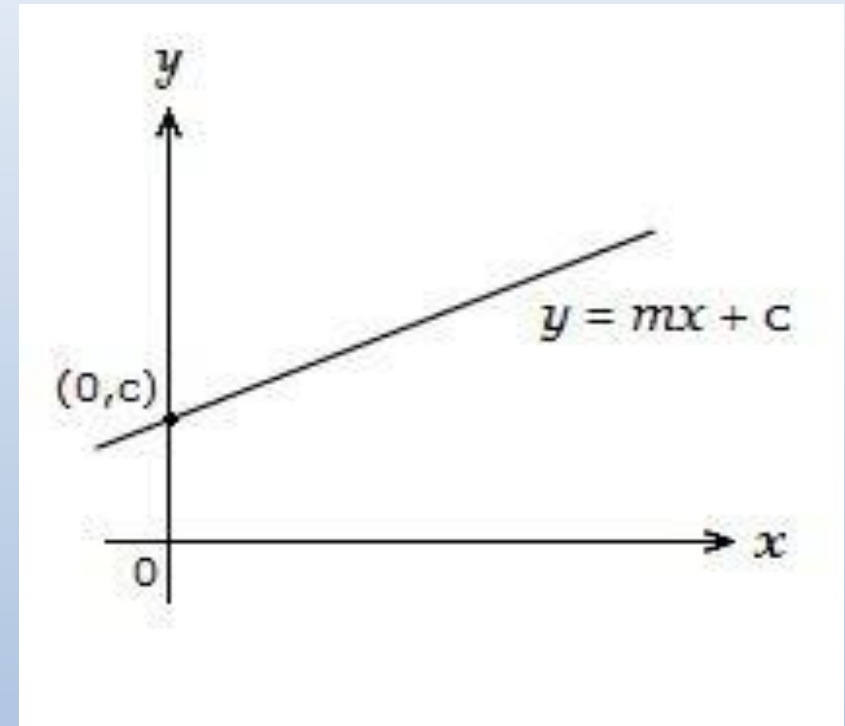
$$\sum(X-X_{\text{mean}})^2 = (1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2 = 10$$

slope (m):

$$m = \text{Numerator} / \text{Denominator} = 20 / 10 = 2$$

intercept (c):

$$c = y_{\text{mean}} - m \cdot X_{\text{mean}} = 7 - 2 \cdot 3 = 1$$



$$\text{Result : } y=2x+1$$

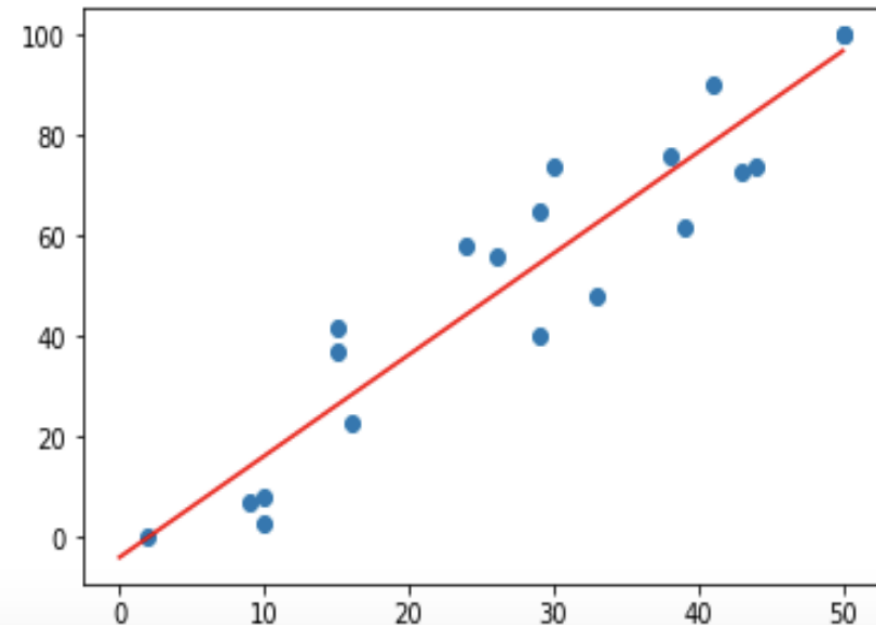
Python Experiment

- Train Linear Regression
- Predict outputs
- Plot regression line

Experiment Link :

```
In [12]: x_lin_reg = range(0, 51)
          y_lin_reg = predict(x_lin_reg)
          plt.scatter(x, y)
          plt.plot(x_lin_reg, y_lin_reg, c = 'r')
```

```
Out[12]: [<matplotlib.lines.Line2D at 0x1a19bff550>]
```



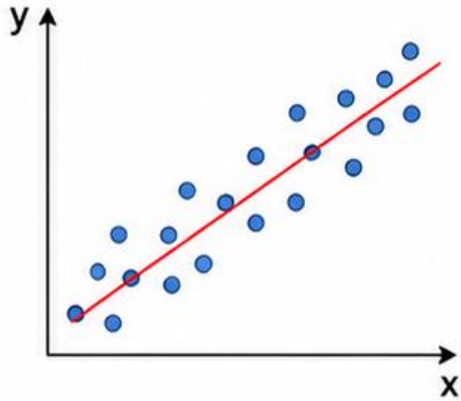
(<https://colab.research.google.com/drive/1zZYCCBKJBiOb43QSX30HpT82B2DmQgMs>)

Regression to ML

STEP 1

LINEAR REGRESSION

Start with a simple linear model to predict outcomes.

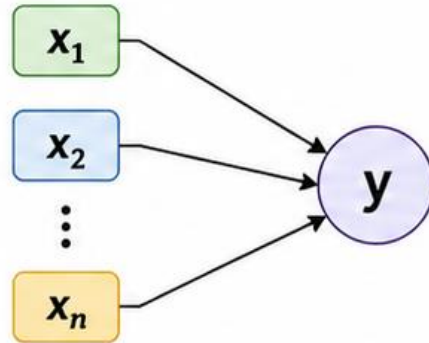


Predicts the relationship between one input (x) and output (y).

STEP 2

MULTIPLE LINEAR REGRESSION

Extend to multiple features (inputs) to improve predictions.



Equation

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c$$

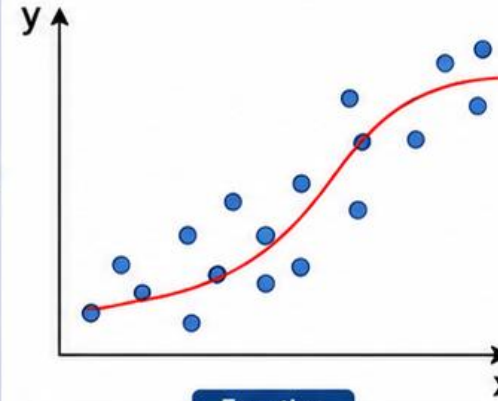


Considers multiple inputs to give better predictions.

STEP 3

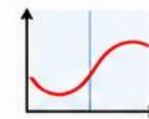
POLYNOMIAL REGRESSION

Introduce non-linearity by adding polynomial terms.



Equation

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$



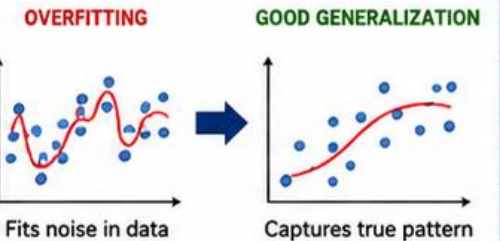
Captures non-linear patterns in the data.

STEP 4

REGULARIZATION

Add techniques like Lasso and Ridge to prevent overfitting.

WHY REGULARIZATION?



RIDGE (L2)



Penalizes large coefficients by squaring them.

LASSO (L1)



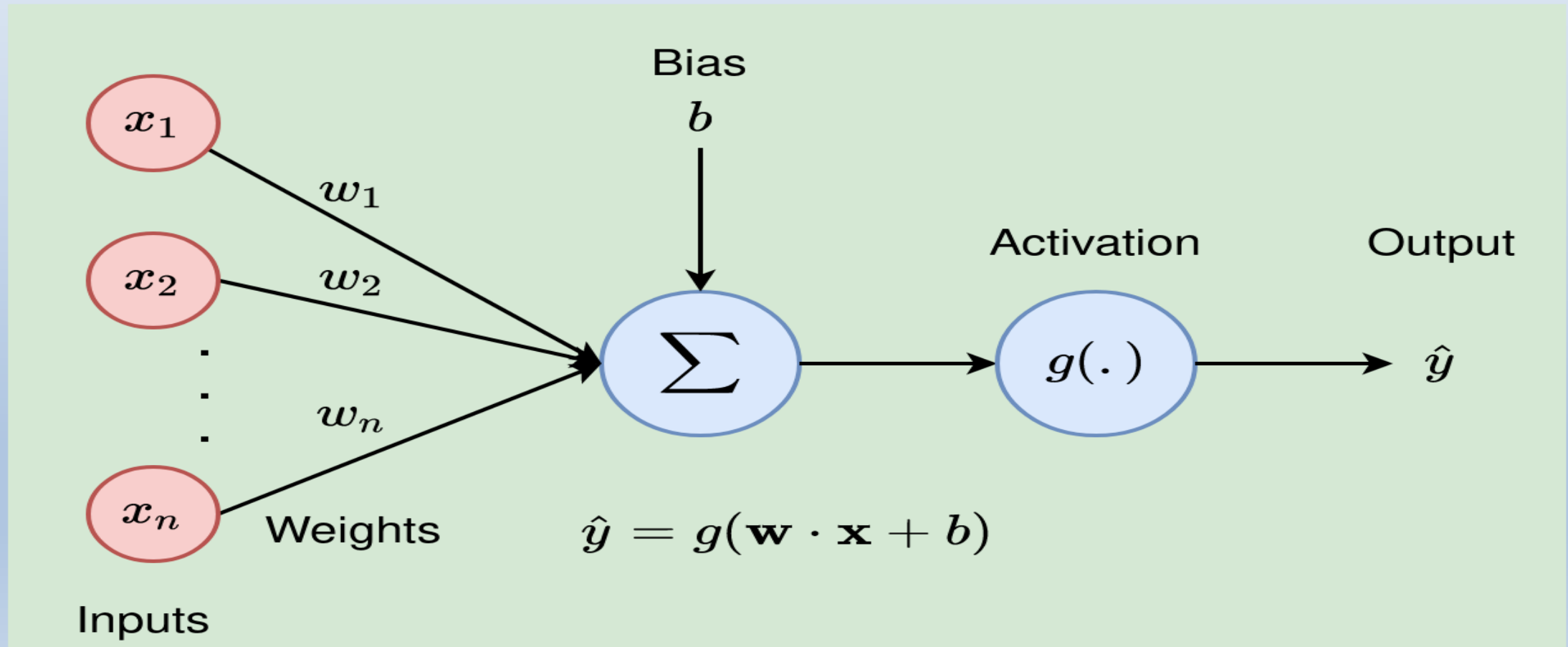
Penalizes large coefficients by absolute value.



Helps prevent overfitting and improves model performance on unseen data.

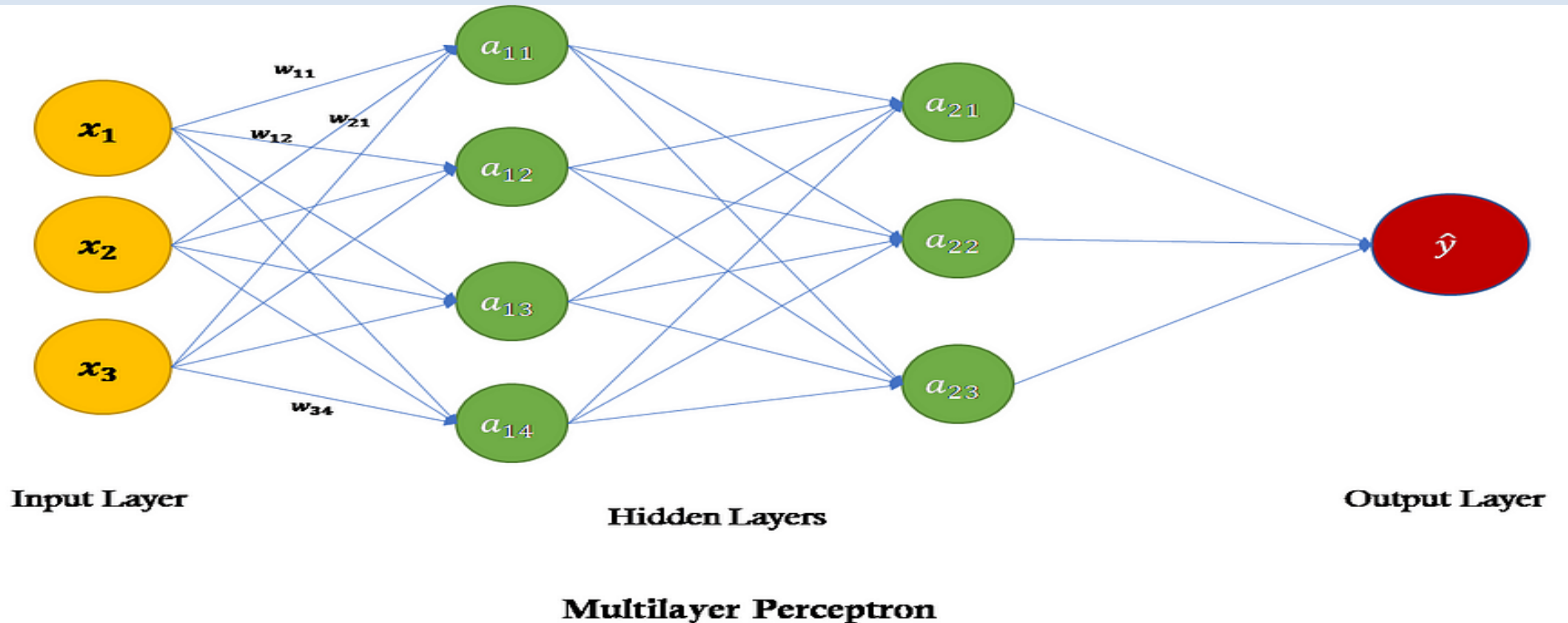
Artificial Neural Networks

Single Neuron (Perceptron):

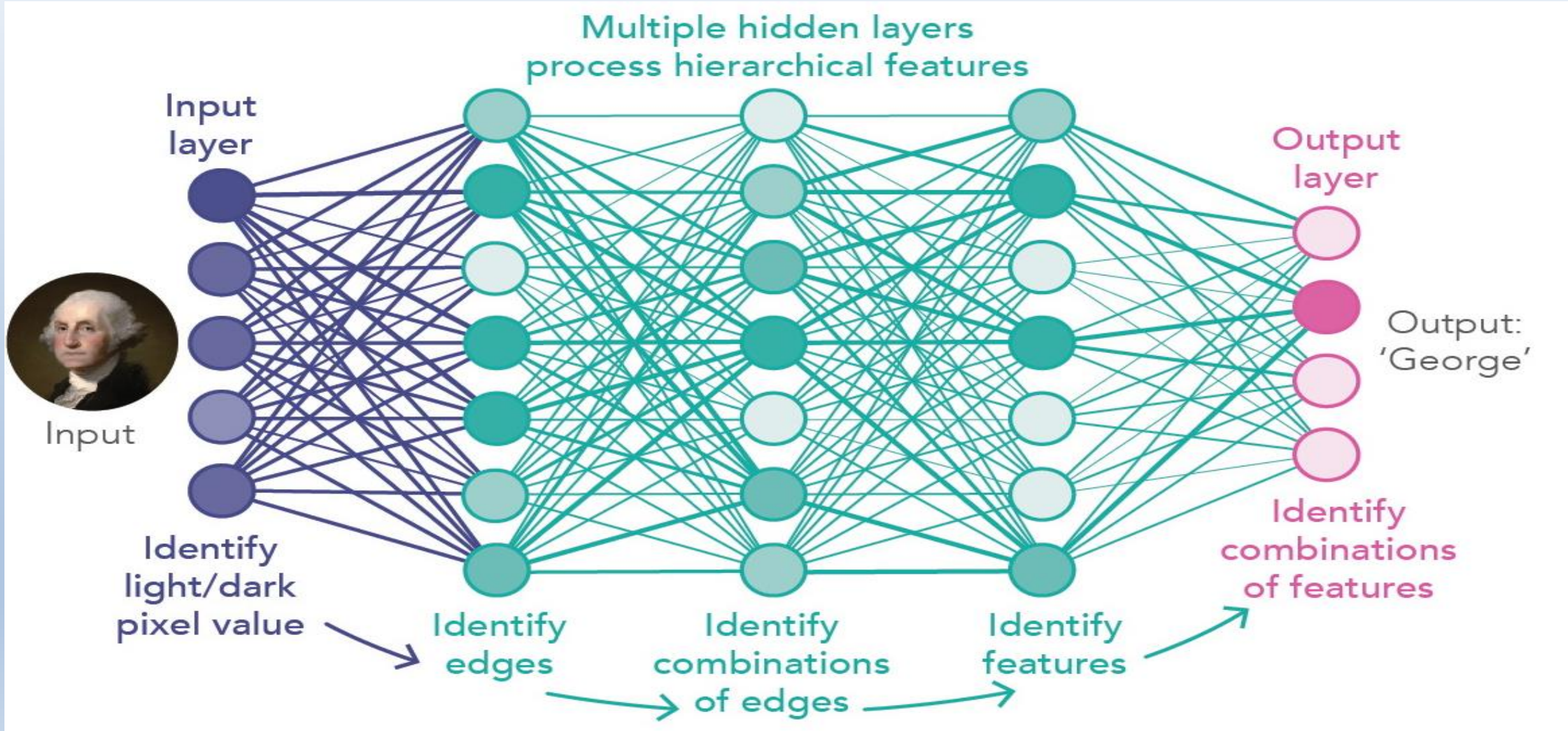


Artificial Neural Networks

Multi-Layer Perceptron (MLP):

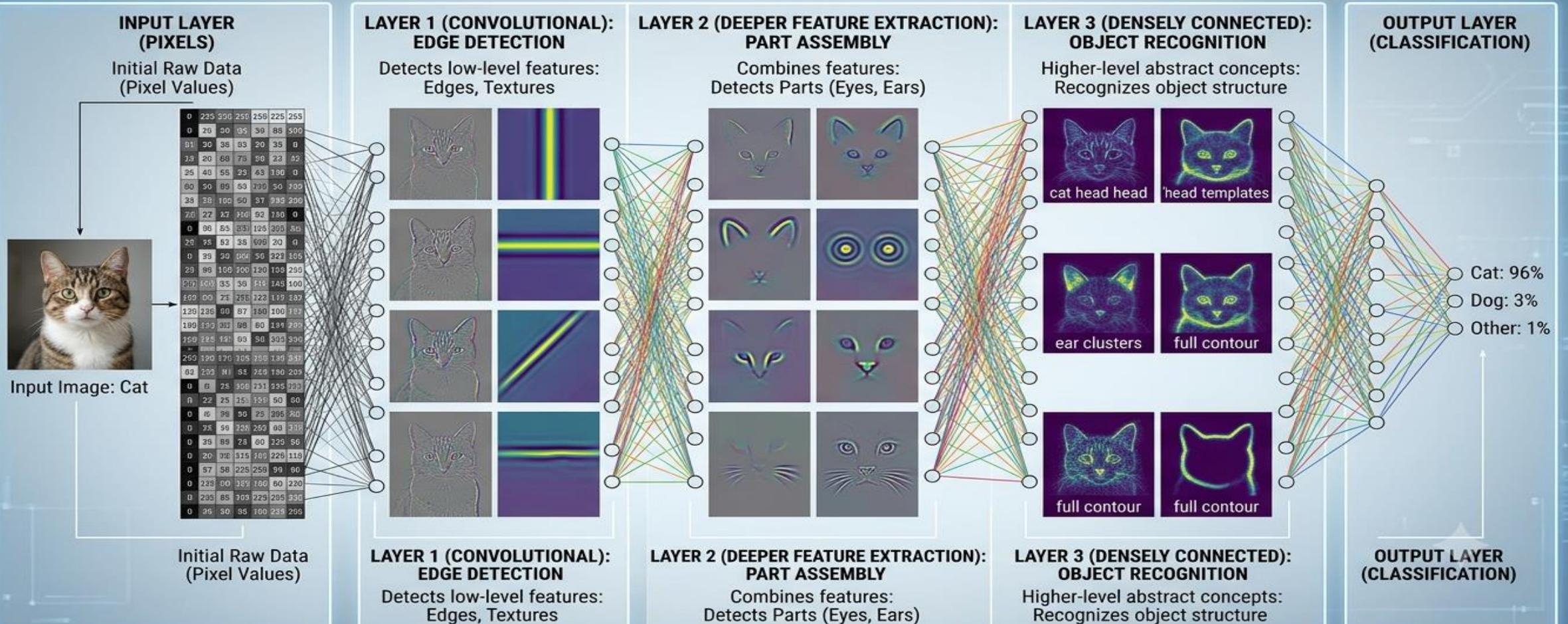


Deep Learning

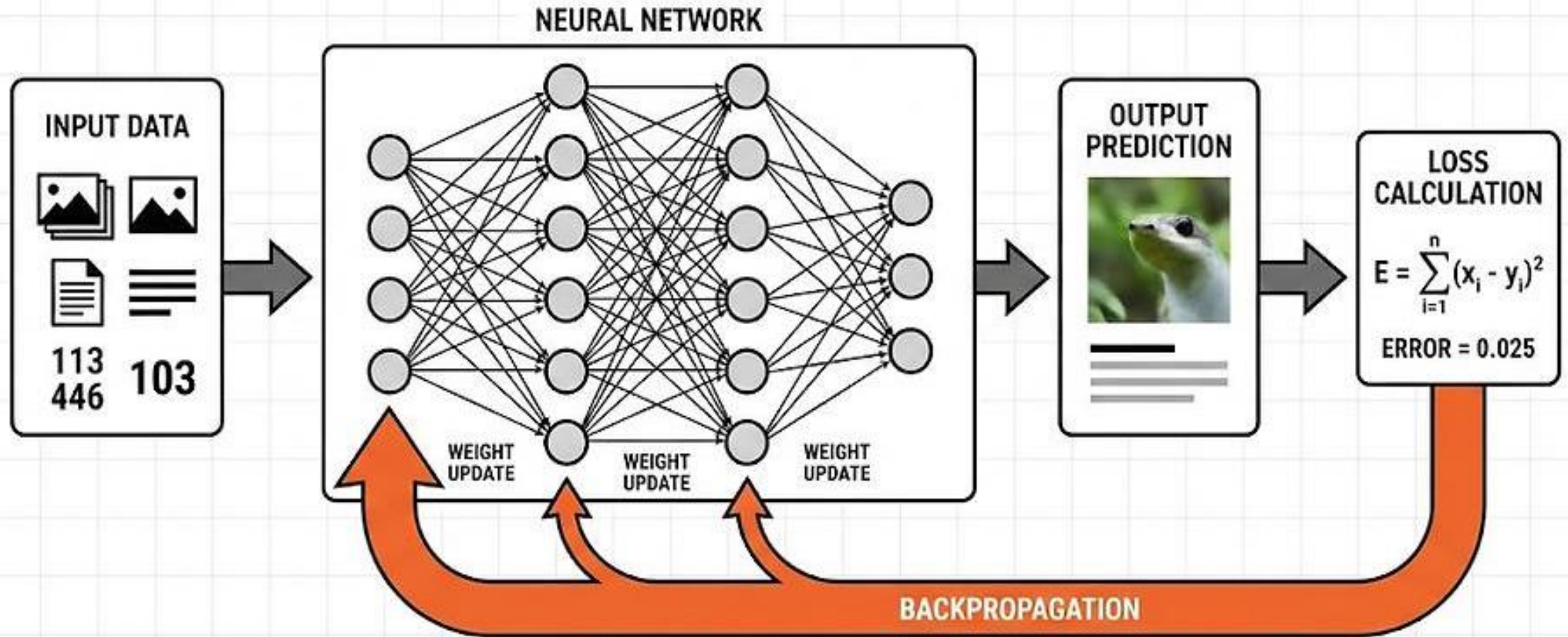


Deep Learning Data Process

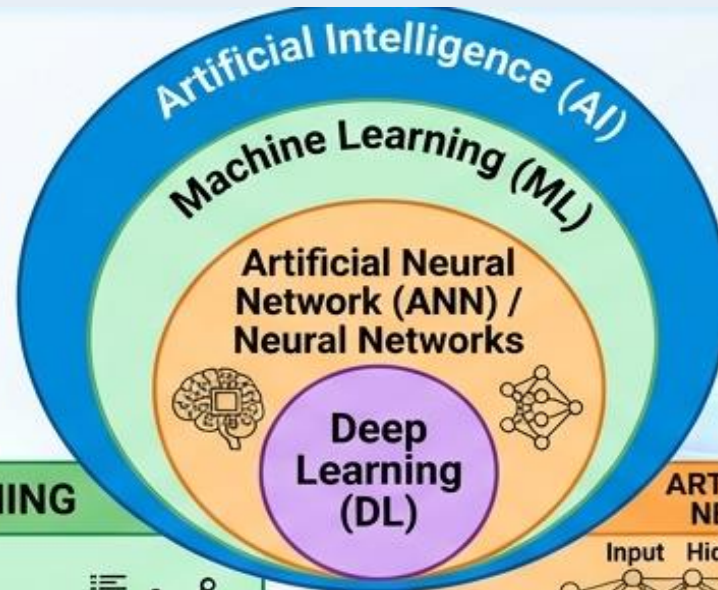
DEEP LEARNING: LAYER-BY-LAYER DATA TRANSFORMATION AND ABSTRACTION

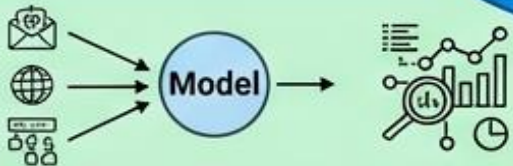
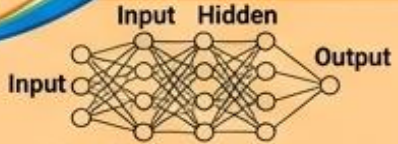
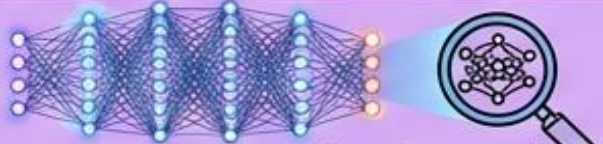


Data Flow of Deep Learning

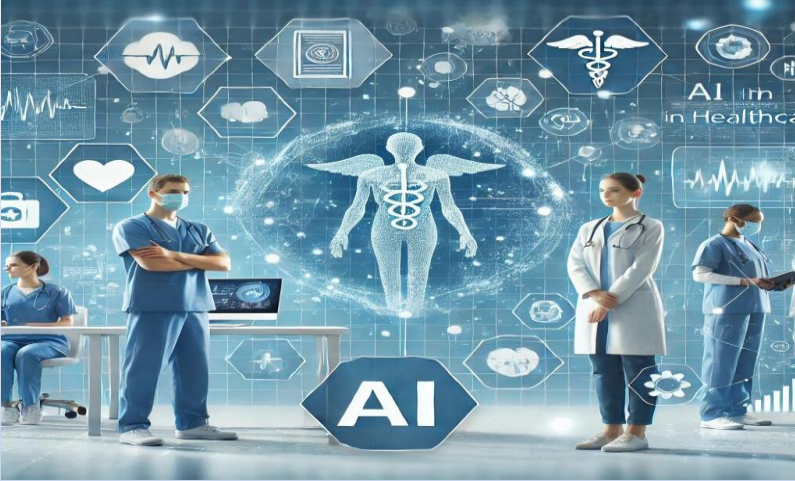


ML vs ANN vs Deep Learning



CHARACTERISTICS	MACHINE LEARNING	ARTIFICIAL NEURAL NETWORKS (ANN)	DEEP LEARNING (DL)
Visual Structure			
Data Requirements	Moderate (e.g., thousands of examples)	More data (e.g., tens of thousands)	Massive (e.g., millions, petabytes) 
Feature Engineering	Manual by domain experts 	Some internal feature discovery, but often pre-engineered	Automatic (Feature extraction learned end-to-end) 
Interpretability Examples	Spam detection, basic recommendation, fraud detection	Pattern recognition, function approximation	Image recognition (e.g., self-driving car visual), natural language processing, complex gaming 

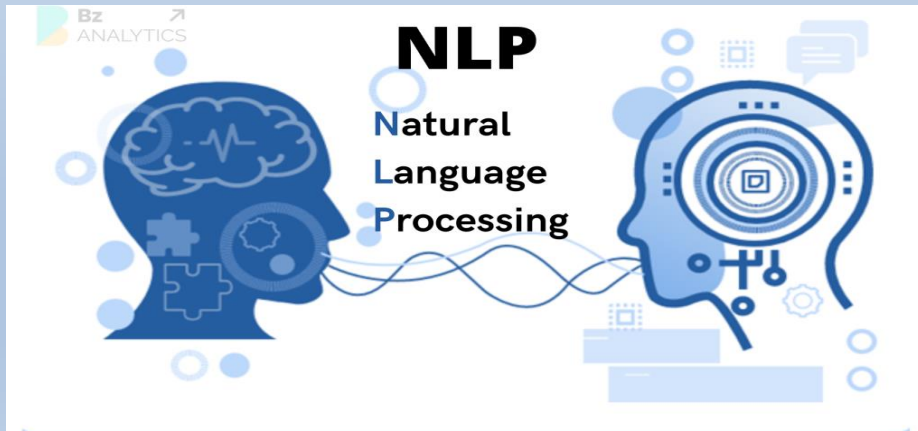
AI Applications



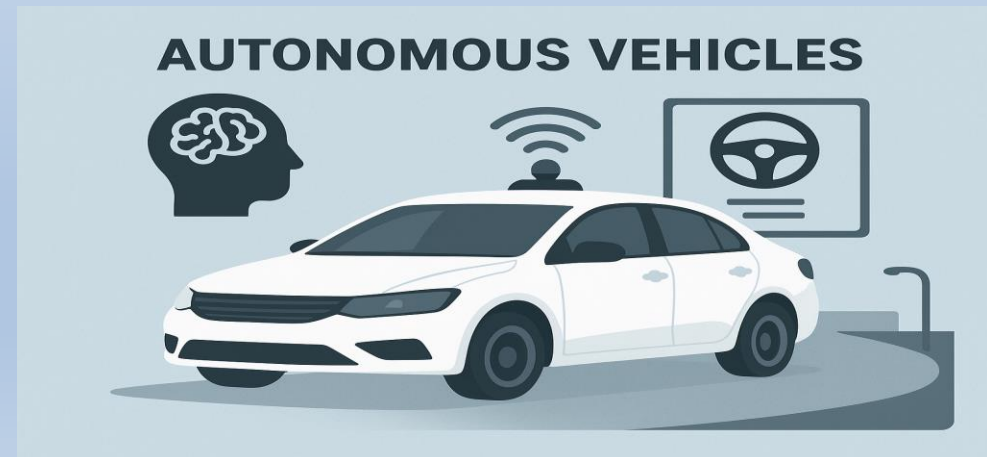
Healthcare



Finance



NLP



Autonomous Vehicles

THANK YOU

Let's we go to Experiment and workspace...

Experiment

1. Shape Finding in AI :

which shape in the image (or) picture. That's findout in using AI prediction program.

[Link : AI_shape_Finding](#)